

Basic Information

Experiment No: 01

Date: 11/09/2026

Title: Bank Account Management System using Classes and Objects

CO: CO1

Objective / Aim

To develop a Java program for a **Bank Account Management System** using classes and objects, demonstrating **deposit, withdrawal, interest calculation, money transfer, inheritance, and method overriding**.

System Requirements

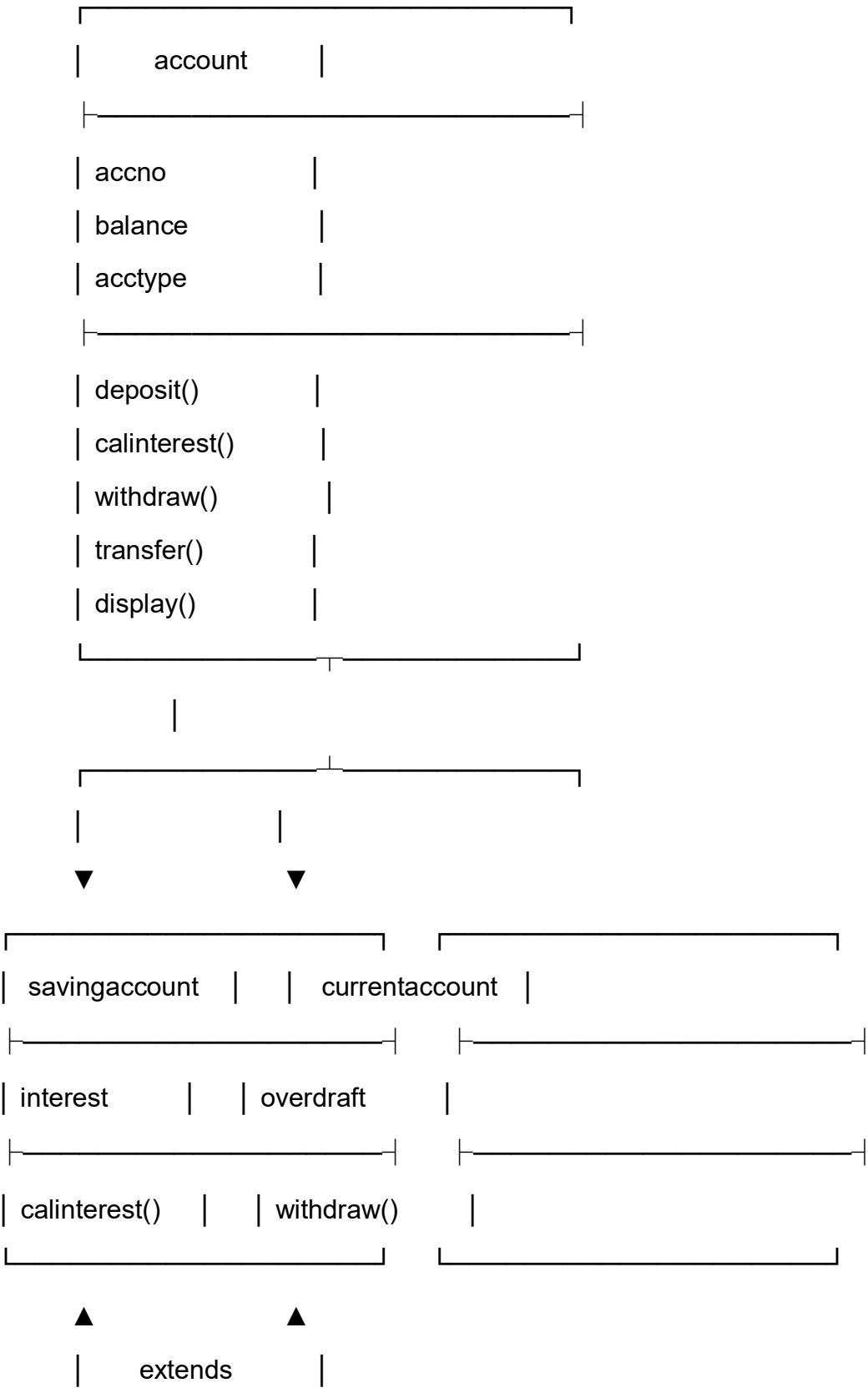
- **Hardware:** Standard PC with Core i3/i5 processor and 4GB/8GB RAM.
- **Software:** Windows/Linux/macOS.
- **Programming Language:** Java.
- **JDK:** JDK 17 or JDK 21.
- **IDE:** Eclipse, IntelliJ IDEA, NetBeans, or VS Code.

Algorithm / Logic Description

1. Create a class account with account number, balance, and account type.
2. Create a constructor to initialize the account details.
3. Define the deposit() method to add money to the account.
4. Define the withdraw() method to withdraw money if sufficient balance is available.
5. Define the transfer() method to transfer money from one account to another.
6. Define the display() method to display account details.
7. Create a savingaccount class that extends account.
8. Add an interest attribute and calculate interest using the calinterest() method.
9. Create a currentaccount class that extends account.
10. Add an overdraft limit to allow withdrawals beyond the available balance.
11. Override the withdraw() method in currentaccount.
12. In the main() method, create objects s1 and c1.
13. Perform deposit, withdrawal, interest calculation, and money transfer operations.

14. Display the final account details.

 UML Class Diagram



_____ account _____

Source Code

```
class account {  
    int accno;  
    int balance;  
    String acctype;  
  
    public account(int accno, int balance, String acctype) {  
        this.accno = accno;  
        this.balance = balance;  
        this.acctype = acctype;  
    }  
  
    void deposit(int amount) {  
        balance = balance + amount;  
        System.out.println("Balance after deposit is: " + balance);  
    }  
  
    void calinterest() {  
    }  
  
    void withdraw(int amount) {  
        if (balance >= amount) {  
            balance = balance - amount;  
            System.out.println("Balance after withdrawal is: " + balance);  
        } else {  
            System.out.println("Insufficient balance");  
        }  
    }  
}
```

```

void transfer(account receiver, int amount) {
    if (balance >= amount) {
        balance = balance - amount;
        receiver.balance = receiver.balance + amount;

        System.out.println("Balance after transfer is: "
            + receiver.balance);
    } else {
        System.out.println("Insufficient balance");
    }
}

```

```

void display() {
    System.out.println(accno);
    System.out.println(acctype);
    System.out.println(balance);
}
}

```

```

class savingaccount extends account {
    double interest;

    public savingaccount(int accno, int balance, double interest) {
        super(accno, balance, "saving");
        this.interest = interest;
    }
}

```

@Override

```
void calinterest() {  
    interest = interest * balance / 100;  
    System.out.println("Interest is: " + interest);  
}  
}
```

```
class currentaccount extends account {  
    int overdraft;  
  
    public currentaccount(int accno, int balance, int overdraft) {  
        super(accno, balance, "current account");  
        this.overdraft = overdraft;  
    }  
}
```

```
// Method overriding  
@Override  
public void withdraw(int amount) {  
    if (amount <= balance + overdraft) {  
        balance -= amount;  
        System.out.println("Amount Withdrawn: " + amount);  
    } else {  
        System.out.println("Overdraft Limit Exceeded");  
    }  
}  
}
```

```
public class BankManagement {  
    public static void main(String[] args) {
```

```
savingaccount s1 =  
    new savingaccount(101, 20000, 5);  
  
currentaccount c1 =  
    new currentaccount(102, 15000, 5000);  
  
System.out.println("Savings Account");  
s1.display();  
  
s1.deposit(5000);  
s1.withdraw(3000);  
s1.calinterest();  
  
System.out.println("\nCurrent Account");  
c1.display();  
  
c1.deposit(2000);  
c1.withdraw(18000);  
  
System.out.println("\nTransfer Money");  
s1.transfer(c1, 4000);  
  
System.out.println("\nFinal Details");  
  
s1.display();  
c1.display();  
}  
}
```

 **Compilation & Execution Commands**

Save the file exactly as:

BankManagement.java

Compile the program:

javac BankManagement.java

Run the program:

java BankManagement

Output Screen

Savings Account

101

saving

20000

Balance after deposit is: 25000

Balance after withdrawal is: 22000

Interest is: 1100.0

Current Account

102

current account

15000

Balance after deposit is: 17000

Amount Withdrawn: 18000

Transfer Money

Balance after transfer is: 18000

Final Details

101

saving

18000

102

current account

18000

Result

The **Bank Account Management System** was successfully implemented using **classes and objects**. The program demonstrates **inheritance, method overriding, deposit, withdrawal, interest calculation, overdraft facility, and account-to-account money transfer**.